

Engle-Granger Two-Step Cointegration Test

Mathematical Notes & Journal

`engle_granger.py` — Dark Wing Project

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1 Introduction: The Core Idea

Imagine you have two stocks, say **Reliance** and **TCS**. Their individual log prices Y_t (Reliance) and X_t (TCS) both wander randomly over time. They are each **non-stationary**, or $I(1)$.

But now ask a different question: is there a number β such that the combination

$$\varepsilon_t = Y_t - \beta \cdot X_t - \alpha \quad (1)$$

does not wander — i.e., ε_t stays close to a fixed mean? If yes, then Y_t and X_t are **cointegrated**. The two prices are attracted to each other by a long-run equilibrium force, even if each one individually drifts.

This is exactly what makes a **pairs trading strategy** work: you bet that the spread ε_t will revert to its mean.

Engle and Granger (1987) proposed a beautifully simple two-step test to verify this relationship, for which they received the **2003 Nobel Prize in Economics**.

2 The Two-Step Procedure

2.1 Step 1: Estimate the Long-Run Relationship via OLS

Run an Ordinary Least Squares (OLS) regression of Y_t on X_t :

$$Y_t = \alpha + \beta \cdot X_t + \varepsilon_t \quad (2)$$

The OLS estimator gives:

$$\hat{\beta} = \frac{\text{Cov}(Y_t, X_t)}{\text{Var}(X_t)} = \frac{\sum_{t=1}^T (Y_t - \bar{Y})(X_t - \bar{X})}{\sum_{t=1}^T (X_t - \bar{X})^2} \quad (3)$$

$$\hat{\alpha} = \bar{Y} - \hat{\beta} \cdot \bar{X} \quad (4)$$

The estimated residuals are:

$$\hat{\varepsilon}_t = Y_t - \hat{\alpha} - \hat{\beta} \cdot X_t \quad \text{for } t = 1, 2, \dots, T \quad (5)$$

Key: If Y_t and X_t are cointegrated, then $\hat{\varepsilon}_t$ should be stationary ($I(0)$).

2.1.1 Matrix Form of OLS

In matrix notation, define $\mathbf{X} = [\mathbf{1}, X]$ (the design matrix with a column of ones for the intercept) and $\mathbf{y} = Y$ (the response vector). Then:

$$\hat{\boldsymbol{\theta}} = \begin{pmatrix} \hat{\alpha} \\ \hat{\beta} \end{pmatrix} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \quad (6)$$

And the residual vector:

$$\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{X} \hat{\boldsymbol{\theta}} \quad (7)$$

In code: `statsmodels.OLS(y, add_constant(x)).fit()` solves (6) and returns `.params = ( $\hat{\alpha}, \hat{\beta}$ )` and `.resid =  $\hat{\varepsilon}_t$` .

2.1.2 Interpretation of $\hat{\beta}$: The Hedge Ratio

$\hat{\beta}$ is called the **hedge ratio**. It tells you:

“For every 1 unit (Rs. 1) long in Y (Reliance), you should be SHORT $\hat{\beta}$ units (Rs. $\hat{\beta}$) of X (TCS) to create a stationary spread.”

Trading implication: If $\hat{\beta} = 0.75$, then the market-neutral pair is:

- Long 1 share of Reliance
- Short 0.75 shares of TCS

The P&L of this portfolio is (approximately) the spread ε_t .

Warning: Standard OLS t -statistics and R^2 are *invalid* when regressing non-stationary series. Do NOT interpret them as usual. Only the residuals $\hat{\varepsilon}_t$ matter.

2.2 Step 2: Test Stationarity of Residuals via ADF

Run the **Augmented Dickey-Fuller (ADF)** test on the estimated residuals $\hat{\varepsilon}_t$:

$$\Delta \hat{\varepsilon}_t = \gamma \hat{\varepsilon}_{t-1} + \sum_{j=1}^k c_j \Delta \hat{\varepsilon}_{t-j} + u_t \quad (8)$$

Hypotheses:

- $H_0 : \gamma = 0$ (unit root $\rightarrow$ residuals are non-stationary $\rightarrow$ NOT cointegrated)
 $H_1 : \gamma < 0$ (stationary $\rightarrow$ residuals are mean-reverting $\rightarrow$ COINTEGRATED)

The test statistic is the t -ratio of $\hat{\gamma}$:

$$\tau_{EG} = \frac{\hat{\gamma}}{\text{SE}(\hat{\gamma})} \quad (9)$$

If τ_{EG} is sufficiently negative (i.e., below the critical value at your chosen significance level), reject H_0 and conclude cointegration.

2.2.1 The Critical Value Problem

Important: Standard ADF critical values (from Dickey-Fuller 1979) are **NOT** correct for Step 2. This is because:

- In Step 1, we *estimated* $\hat{\beta}$ from the same data.
- The OLS estimator selects $\hat{\beta}$ to minimise the residual variance.
- This biases the residuals toward looking more stationary than they really are.

Solution: Use **Engle-Granger specific critical values** (from MacKinnon, 1991, 2010). These are more negative than standard ADF critical values to account for the pre-estimation bias.

`statsmodels.adfuller()` automatically applies MacKinnon (2010) response surface critical values, which are correct for the Engle-Granger context.

3 The Spread as a Trading Signal

If the pair is cointegrated, the estimated spread $\hat{\varepsilon}_t$ from (5) is a stationary process. We can model its behaviour with the **z-score**:

$$z_t = \frac{\hat{\varepsilon}_t - \mu_\varepsilon}{\sigma_\varepsilon} \quad (10)$$

where $\mu_\varepsilon = \mathbb{E}[\hat{\varepsilon}_t]$ and $\sigma_\varepsilon = \text{std}(\hat{\varepsilon}_t)$.

Trading rules:

Condition	Action	Reason
$z_t > +2$	Short spread: Short Y , Long $\beta \cdot X$	Spread too wide above mean
$z_t < -2$	Long spread: Long Y , Short $\beta \cdot X$	Spread too wide below mean
$ z_t < 0.5$	Exit position	Spread reverted to mean

The threshold of ± 2 standard deviations comes from the normal distribution (95.4% of observations lie within $\pm 2\sigma$ for a Gaussian).

4 Assumptions and Limitations

4.1 Assumptions

1. Both Y_t and X_t are $I(1)$ (non-stationary in levels, stationary in differences).
2. The cointegrating vector is unique (only one long-run equilibrium).
3. The error ε_t is serially uncorrelated after augmenting with enough lags.

4.2 Limitations

1. **Single vector:** Engle-Granger can only detect one cointegrating relationship. For $m \geq 3$ series, use the Johansen test (`johansen.py`) instead.
2. **Direction matters:** The hedge ratio $\hat{\beta}$ from regressing Y on X is NOT the same as regressing X on Y . In practice, choose the series that *adjusts more slowly* as the dependent variable.
3. **Regime changes:** $\hat{\beta}$ is fixed over the full sample. Use `rolling_cointegration.py` and the Kalman filter to track time-varying β .
4. **Two-step bias:** Estimation error from Step 1 propagates into Step 2. This is called the “generated regressor” problem.

5 SuperConsistency of the OLS Estimator

Even though OLS is unreliable for non-stationary series in general, it is **superconsistent** when cointegration truly exists.

Normally, OLS converges at rate $1/\sqrt{T}$ (i.e., needs many observations to be precise). In the cointegrated case, OLS converges at rate $1/T$ — **much faster**.

$$\hat{\beta} - \beta_{\text{true}} = O_p\left(\frac{1}{T}\right) \quad (\text{superconsistent}) \quad (11)$$

This means even with moderate sample sizes, the OLS estimate of the hedge ratio is very accurate if the pair is truly cointegrated.

6 Connection to Dark Wing Pipeline

Step 1: Unit Root: Use `StationarityTest.check_stationarity()` to verify both Y_t and X_t are $I(1)$.

Step 2: OLS: `EngleGrangerTest._run_ols()` estimates $\hat{\alpha}$ and $\hat{\beta}$ from (3).

Step 3: ADF on Residuals: `EngleGrangerTest._run_adf_on_residuals()` tests if $\hat{\varepsilon}_t \sim I(0)$.

Step 4: Decision: If $p\text{-value} < 0.05$, pair is cointegrated.

Step 5: Spread: `EngleGrangerTest.get_spread()` returns $\hat{\varepsilon}_t$ for z-scoring and signal generation.

Step 6: Kalman: The static $\hat{\beta}$ is replaced by a dynamic, time-varying β_t from `dynamic_beta.py` for live trading.

7 Numerical Example

Data: Reliance and TCS log prices, 2020–2024 (daily, 1000 observations).

OLS Regression: $\ln(REL_t) = \hat{\alpha} + \hat{\beta} \cdot \ln(TCS_t) + \hat{\varepsilon}_t$

$$\hat{\alpha} = -2.1073$$

$$\hat{\beta} = +0.8421$$

Spread: $\hat{\varepsilon}_t = \ln(REL_t) - 0.8421 \cdot \ln(TCS_t) + 2.1073$

ADF test on $\hat{\varepsilon}_t$:

ADF Statistic	−3.84
$p\text{-value}$	0.0027
Critical value (5%)	−3.41
Decision	Reject H_0 -> Cointegrated (yes)

Spread statistics:

$$\mu_{\varepsilon} = 0.0000 \quad (\text{by construction})$$

$$\sigma_{\varepsilon} = 0.0483$$

Trading signal at current time t : If spread = 0.1015, then $z_t = 0.1015/0.0483 \approx 2.10 > 2.0$.

$\Rightarrow$ **Short the spread:** Short 1 unit Reliance, Long 0.8421 units TCS.